

- 25 Butanone reacts in a three-stage process to produce 2-methylbut-2-enoic acid, $\text{CH}_3\text{CHC}(\text{CH}_3)\text{COOH}$.

Which reagents could be used for the three stages?

	first stage	second stage	third stage
A	$\text{H}_2\text{SO}_4(\text{conc}) / \text{heat}$	HCN	$\text{NaOH}(\text{aq}) / \text{heat}$
B	$\text{NaOH}(\text{aq}) / \text{heat}$	HCN	$\text{H}_2\text{SO}_4(\text{conc}) / \text{heat}$
C	HCN	$\text{H}_2\text{SO}_4(\text{conc}) / \text{heat}$	$\text{NaOH}(\text{aq}) / \text{heat}$
D	HCN	$\text{NaOH}(\text{aq}) / \text{heat}$	$\text{H}_2\text{SO}_4(\text{conc}) / \text{heat}$

- 26 K_a values for two acids are given